Practical Business Analytics COM3032 – COMM074

Coursework Deadline Wednesday 20th of May 2026 at 16:00

Submit before the deadline - a submission of 16:00 is considered late by Surrey Learn and penalties are added automatically.

Module Aims

Your learning objectives for this module are based on the Module Descriptor, which are:

- To deploy business analytics pipeline on a real-world business dataset.
- To create a functioning Python code using Jupyter Notebooks and\or Google Colab.
- To make appropriate judgments on what Python libraries, and data analytics techniques to use.
- To extract high-level information and insights derived from low-level (raw) data.
- To design appropriate visualisations to help present and communicate the discovered insights to decision makers in a form of actionable recommendations.
- To learn how to work in a team composed of people with diverse background and skills (and personalities).

Main Objectives of the Assessment

The aim of this assignment is to generate value and insight from processing heterogeneous data. This will be achieved by implementing several analytic methods / techniques / algorithms, evaluating them, and comparing the effectiveness of the adopted approaches.

Assessment Overview

The coursework is 80% of the module assessment and is group based. As a group you need to select an open-source dataset\s with a business application and apply business analytics to this problem based on the topics covered in the lectures and labs. You must write code in Python using Jupyter Notebooks that read in, process, analyse and visualise the data chosen for this assessment. You must also prepare a business report that introduces the problem and its relevance to a business or industry, details your data, explains the process and methods used in code, presents the results (from your code) in a coherent way and discusses the results relative to your chosen problem. A full marking scheme is at the end of this brief.

The final report should be an original and Group submission, but it will be underpinned by a group effort of effective sharing of data and partial results. It is important that individual contributions shared among the group are clearly defined (see “Authorship Contribution Statement” below for further details). These contributions should be agreed upfront in a designated meeting.

Deliverables

As a group, you will create and deliver:

1- Business Analytics Report (1 Pdf file per Group)

Your group will produce a concise, non-academic Business Analytics Report. The report should be succinctly written, making use of tables and charts where appropriate. You are to:

a) Provide a problem definition and objectives. It will contain an overview of your chosen dataset, including a data dictionary.

b) Explore your dataset and explain your choice of data preparation approaches as well as each of the steps. You need to make sure you justify your choices adequately.

c) Document how you choose appropriate machine learning algorithms and evaluation methodology with choice of metrics, justify your approach, along with your assumptions.

d) Provide the summarised results from your Python code in tables/charts suitable for business audience. This must include
(1) appropriate data exploration and pre-processing steps,
(2) suitable modelling techniques,
(3) appropriate evaluation of your models, and
(4) your results, visualisation, conclusions and recommendations for "senior management".

There is no page limit, but the report should not include 100 pages of meaningless figures.

2- The dataset/s used for the analysis.

3- Python Code used in the assessment

Your group is required to write multiple Python Notebooks to meet the project objectives that are set in the **Introduction Section** of the report.

- The first Python Notebook must load and process the dataset, undertake your data cleaning steps, perform exploration and preparation (pre-processing) tasks.
- 1 individual Python Notebook for each member should be used to read in the pre-processed data and to train the models using chosen ML algorithms, evaluate and output results, and provide evaluation visualisation as appropriate:
    - The Python code should be properly commented to explain the steps and overall process.
    - Source code is provided on Surrey Learn for each individual lab that gives you example implementations in Python. You are welcome to use any/all/none of this code but must mark in your code where you do so.
    - You will NOT get a good mark if you only demonstrate a minimum level of skill by extensively use Python code copied from other sources. In general, when in doubt, provide a simple reference (at least as a courtesy) where you have taken and used any code from an external source – even if you have amended this. You will need to judge if you believe the code to be "sufficiently different"

to the source so that you do not need to reference. If the code is generic or "obvious" then you may consider this unnecessary. Remember to include comments to document your Python code and choice of libraries. You should comment your own functions following the recommended approach in the labs.

- Your Python code should output [text] tables and any plots/graphs/visualisations as necessary. These can be exported and used in your Business Analytics Report and presentation where appropriate.

Deliverables Deadlines

The deadlines for the above deliverables are listed below:

Deliverables	Deadline
Groups formation	Week 3 (Friday 20th of February) Email module leader the names of group members.
Business Analytics Report	Wednesday 20th of May at 16.00
Dataset/s	Wednesday 20th of May at 16.00
Python codes used to meet the aim and objectives of the assessment	Wednesday 20th of May at 16.00

Detailed Marking Scheme

Report 80%					
	Title Page (Project Title, Group Name and Group Members)				
	Section	**Group OR Individual**	**Details**	**Weight**	**Marking Criteria**
1	**Introduction:** – Business Understanding – Data Understanding – Setting Research Question (Business Objective\s)	Group Work (Marked as a group effort)	• A background to the general problem and why it is important (e.g., what the stock market is and why it's important for business. • You must give a clear problem statement with objectives, based on your dataset, including any assumptions. • In other words, what problem are you trying to solve (e.g., predicting future price) and why this is important for business analytics. • What are the major questions you wish to answer? • What techniques do you plan to use? • What data are you looking at and where is it from • Overview of the data (size, number of features, features' data type, etc) • You must make sure that the dataset (possibly from combined sources) can be used to answer the questions/hypotheses. • You need to make use of CRISP-DM approach; describe your approach in terms of the steps taken specific to your project.	15%	**Dataset/s Relevance, Complexity and Justification:** • The problem/research question, project objectives, and chosen techniques should be relevant to the chosen dataset/s. • The chosen dataset must align with the task requirements and business goals. Assess how well the dataset is suited to answer the analytical questions and provide insights relevant to the task. • The complexity of the dataset in terms of the number of features, size of the dataset, and the range of values in each feature. A more complex dataset should be rewarded for the challenge it poses. • The rationale provided by the students for selecting the dataset. Ensure it aligns with the goals of the task and the potential insights

			You must demonstrate your understanding of the chosen dataset, in context of your stated objectives, through proper data inspection. This can include: • Dataset loading: You must load the chosen dataset(s) into the Python Notebook. • Data inspection: You must check the data types of the variables in the dataset and determine if a field/variable is a numeric or symbolic (e.g., categorical), etc. • Explain how knowing the data types will influence your decisions related to data pre-processing. • Hypothesize which fields from the dataset will be included in your final data sample. • What are the inclusion criteria? • Similarly, hypothesize the fields that will be excluded from your project and decide and justify the exclusion criteria. These tasks are typically performed to ensure the validity of the data and to gain an initial understanding.		that can be derived from the chosen data.
2	**Data Pre-processing and Exploration** – **Data Exploration** – **&** – **Data Preparation and Cleaning**	Group Work (Marked as a group effort)	In general, this section focuses on exploring and preparing the data to be in a form suitable for subsequent modelling and evaluation. More particularly, you need to perform detailed data preparation and cleaning steps. This will require: • Dataset processing: The steps required to analyse each field/record [such as handling missing data and outliers] and how you	15%	**<u>Data Cleaning:</u>** • How effectively the student handled missing data, outliers, and inconsistencies. Deduct points for inadequate or improper handling. **<u>Feature Engineering:</u>** • The creativity and effectiveness of feature engineering. Reward for

	For this section, you (as a group) should produce **one Python Notebook** that reads-in the original dataset and perform all the data understanding, preparation and exploration steps, and eventually create the pre-processed dataset that is ready for modelling		might combine features or measure the quality of the dataset. • Perform necessary data transformation normalisation, scaling, encoding, feature selection and constructions. Sample tasks involve some/all of the following: ✓ Field transformation ✓ Discretisation ✓ Dimensionality reduction ✓ Combined different datasets. ✓ Derivation of new fields ✓ Field encoding ✓ Normalisation ✓ Class balancing Also, you need to have detailed data exploration and visualisations. This will require generate: • Detailed summary statistics • Graphical analysis such as: ✓ Histogram ✓ Density plot ✓ Boxplot ✓ PCA ✓ etc Finally, you should produce a summary table, with the final selected input fields to the model(s) and an analysis/description of each. You will need to justify and explain your chosen approach to each step of data-preparation in context of the chosen ML algorithms		creating new informative features or transforming existing ones. **Scaling/Normalization:** • The student appropriately scaled or normalized the features, making sure they understand the importance of this step for various machine learning algorithms. **Data Split:** • The student correctly split the dataset into training, validation, and test sets, ensuring they understand the significance of proper data splitting for model evaluation.

3	**Modelling** – **Machine learning Algorithms Implementation** For this section, each member should produce **one Python Notebook** file, which contains Python code that reads-in the pre-processed dataset created in Section 2 and perform all the modelling and evaluation steps	Individual Work (Marked individually)	Each member of the group is expected to implement and apply two supervised ML techniques to perform supervised learning. You are required to train two models using ML algorithms. The models and evaluation must be implemented in your Python Code. One of the models should be unique compared to rest of the group members, and the other model can be the same, however, with different implementation parameters to enable comparison. Please remember that CRISP-DM is an iterative project approach, this stage typically requires repeated iterations of modelling that will involve: • The selection of ML modelling algorithms, e.g., choose appropriate algorithms from available Python libraries. • The choice of the ML algorithm parameters/hyper-parameters, e.g., to reduce overfitting and optimise performance you might try a range of values (tuning) • The training and evaluation of the ML model(s) using the pre-processed dataset. The design of the modelling tasks must be sound and justified, and each ML modelling algorithm discussed along with its strengths and weaknesses in context of this real-world dataset. Appropriate parameter settings must	20%	**Model Selection:** • The appropriateness of the chosen models for the given task, considering factors such as complexity, interpretability, and suitability for the problem. **Hyperparameter Tuning:** • The effectiveness of hyperparameter tuning, considering how well the student improved model performance through this process. **Model Training:** • The student's ability to correctly use the training dataset and the correct set of ML models' parameters to train the models.

			be used, stated and justified. Inevitably, there will be elements of experimental trial and error.		
4	**Performance evaluation** – **Comparison of Models' performance** Use the same Python Notebook used in Section 3.	Individual Work (Marked individually)	• For both ML techniques, each member of the group should produce their own results (Evaluation Metrics), • Distribute/exchange their results among all the other members. • Please note that the Evaluation Metrics should be the same for all group members to enable comparison. • Each member is expected to independently compare, discuss and evaluate these shared results. The robust evaluation of the modelling results is critical to success of a Business Analytics project. You must have a clear evaluation approach that is described and justified. The resulting models in your Python code must each be evaluated and compared for their ability to generalise on unseen data using your evaluation approach.	15%	**Model Evaluation:** • The student's ability to choose the relevant evaluation metrics based on the set requirements to evaluate the models, ensuring they understand the fundamentals of model training and evaluation.
5	**Discussion, Conclusion and Recommendations** – **Discussion of the findings,** – **Conclusion** – **Recommendations**	Group Work (Marked as a group effort)	• How did your model do? (marks (grades) are NOT correlated with model performance) • How have you analysed the performance and what does this indicate about the data you are looking at? • Visualisation (graphs and\or dashboard, etc.) (There is no page limit, but the report should not include 100 pages of meaningless figures)	15%	**Interpretation:** • The depth and relevance of the insights drawn from the model's performance and how well the student connects these insights to business implications using appropriate visualisation choices.

			Results must be compiled from the output of your Python Code. These must be presented along with visualisation, conclusions and recommendations in a written report suitable for a business audience. You must analyse and discuss these results, and your interpretation may confirm or reject any initial hypothesis or indicate newly discovered trends/patterns in the data. This is an important part of your Business Analytics project – to be able to clearly communicate your results to a non-technical business audience.		**Conclusion and Recommendation:** • The clarity and completeness of the conclusion summarizing the findings of the analysis. • The quality and feasibility of the recommendations provided for the business based on the analysis.
6	**Author Contribution statement**	Group Work (Marked as a group effort)	• AUTHORSHIP CONTRIBUTION: Authors make substantial contributions to conception and design, and/or acquisition of data, and/or analysis and interpretation of data and/or evaluation, and/or visualisation. • Each group member should rate their contribution (%) to the group-based sections. • Individual Signatures to authenticate group members' statements and ratings.	Pass/Fail This section is not weighted but it is important to decide the group and individual grades	

Code 20%		
Group OR Individual	**Details**	**Marking Criteria**
Group AND Individual	• Submitted as a single zip file containing the all Python Notebooks. • Each section must be run with associated output shown. • The code should be well mapped to the group and individual sections in the report. PLEASE NOTE: If you choose to use and/or adapt code snippets from anywhere else (e.g., GitHub, Kaggle, Gen AI etc.), you must acknowledge this and provide the complete reference, to avoid any issues with plagiarism.	Marking criteria for data science code can vary depending on the specific context, the level of expertise expected, and the goals of the project. However, here are some general criteria that can be used to evaluate data science code: • <u>**Code Structure and Organization:**</u> **Modularity and Readability:** Is the code well-organized into functions or modules, making it easy to read and understand? **Comments and Documentation:** Are there meaningful comments and documentation explaining the purpose and functionality of the code, including variable names and functions? • <u>**Data Preparation:**</u> **Data Cleaning:** Has the data been thoroughly cleaned, including handling missing values, outliers, and inconsistencies? **Data Transformation:** Are the necessary transformations applied to prepare the data for analysis (e.g., normalization, encoding categorical variables)? • <u>**Exploratory Data Analysis (EDA):**</u> **Insights and Visualizations:** Are appropriate exploratory data analysis techniques used to gain insights into the data, and are the results effectively visualized? **Statistical Analysis:** Does the code include relevant statistical analysis to summarize and describe the data? • <u>**Feature Engineering:**</u> **Feature Creation:** Are new features derived from the original data to enhance predictive power or insights? b. **Feature Selection:** Is there evidence of feature selection techniques to choose the most relevant features for the analysis? • <u>**Modelling**</u>: **Model Selection:** Have appropriate models been chosen based on the problem and data? **Model Training:** Is the model trained using a suitable approach, and are hyperparameters tuned effectively?

		• **<u>Model Evaluation and Performance:</u>** **Evaluation Metrics:** Are appropriate evaluation metrics used to assess the model's performance (e.g., accuracy, precision, recall, F1-score, ROC-AUC)? **Comparisons:** Are multiple models compared, and is there a clear justification for choosing a particular model? • **<u>Data Visualization and Interpretability:</u>** **Visualizations:** Are meaningful and informative visualizations created to present the analysis and results effectively? **Interpretability:** Is there an effort to explain the model's decisions and make the analysis interpretable to non-technical stakeholders?

Coursework Structure

	Section Name	Code	Group or Individual	Code File
1	Introduction (Business Understanding)	Basic code for basic data inspection	Group	1 Python Notebook per GROUP
2	Data Understanding, Pre-Processing and Exploration	Detailed Code for data preparation and Exploration	Group	
3	Modelling	Detailed code for models' development and tunning	Individual	1 Python Notebook per STUDENT
4	Performance Evaluation	Detailed code for models' performance evaluation	Individual	
5	Discussion, Conclusion and Recommendations	Detailed code to create visualisation (graphs and\or dashboard) to help decision makers understand the important insights.	Group	1 Python Notebook per GROUP
6	Author Contribution Statement		Individual	

University Policy on the Use of Generative AI

The University is aware that there is a high-profile discussion on the use of generative AI tools, such as ChatGPT and Gemini, and you may have already heard of or have even used them in your learning and assessments. Please be aware that the unethical and irresponsible use of such tools in your assessments could risk breaching the University regulations and be identified as academic misconduct. The information the generative AI tools present is not always trustworthy or accurate. Therefore, critically evaluating the information you have retrieved from such tools is very important in your assessments. To maintain academic integrity, please make sure that you always submit your own work based on your own learning with appropriate referencing where necessary. Please refer to the student guide on Academic Integrity and use of Generative AI tools.

Acceptable Use of Generative AI:

- **Data Cleaning Assistance:** Students can use AI tools to suggest methods for cleaning datasets (e.g., handling missing values or detecting outliers). However, they should apply these methods themselves and justify their choices in the coursework.
- **Brainstorming and Suggestions:** AI can help students suggest machine learning models to use and hyperparameters tunning techniques. The students should then experiment with these models, analyse the results, and explain their findings and decisions on their own.

Unacceptable Use of Generative AI:

- **Automated Model Development:** Submitting models entirely generated and tuned by AI without understanding the underlying algorithms or making decisions based on the data characteristics.
- **Lifecycle Documentation:** Allowing AI to document the entire data science lifecycle without personal insights or reflections on the process, challenges, and learnings.

- **Directly Generating the Report:** Do not use Generative AI to write the entire report or any sections of it. The report should be your own original work, demonstrating your understanding of the concepts and your ability to apply them to the given dataset.
- **Unattributed Use of External Code from AI Sources:** If AI suggests code from external sources (like GitHub), you must provide references. Using external code without proper citation or modification, especially code from Generative AI, can lead to issues with plagiarism.
- **Direct Copying of AI-Generated Code:** Relying on AI to generate the core Python scripts used in the analysis. The coursework requires that you write original code. Copying AI-generated code without adaptation and understanding is considered plagiarism.

Overall, the coursework requires that you write original code, thus, make sure not to copy and paste code or text generated by Generative AI. If you use any AI-generated content, clearly indicate its source and explain how you have adapted or utilized it in your work.